

Project Briefs

AI-powered return-fraud detection for Indian D2C e-commerce brands

Problem Statement

E-commerce platforms in India face return fraud rates far higher than global averages, driven heavily by Cash-on-Delivery (COD) orders where customers order, refuse delivery, or return items after use with almost no verification at the point of purchase. D2C brands and mid-market sellers lose heavily to wardrobing (wearing and returning clothes after a wedding or festival season), empty-box returns, and serial over-ordering (ordering multiple sizes with intent to keep one and return the rest). Enterprise fraud tools are priced for large players, leaving smaller D2C brands with no affordable way to fight this.

Solution

A lightweight application for Indian D2C brands that analyzes each return or refusal risk in real time. It factors in COD refusal history, return frequency per customer, address-mismatch patterns, and seasonal wardrobing signals (e.g. ethnic wear returned right after wedding season), using the merchant's own order data. Low-risk returns get auto-approved instantly to keep customer trust high; high-risk ones get flagged for manual check or a small verification step such as photo proof or video unboxing requirement. Pricing is a small percentage of fraud losses actually prevented, making it a zero-upfront-risk pilot for a small D2C brand to try.

Solution Overview

A simple standalone application that receives a merchant's order and return data, looks at each customer's order and return history, and gives every return request a risk score in real time. Low-risk returns get approved automatically. High-risk cases (repeat over-orderers, COD refusal patterns, returns right after festival season) get flagged for a quick manual check or a photo/video proof step before refund.

Target Users & Problem Fit

Small-to-mid Indian D2C brands (\$50K-\$5M revenue range) selling through their own e-commerce operations with COD checkout. They lose real money to return fraud every month but cannot afford enterprise fraud tools built for large retailers. They will use it because it is cheap to try (pay only a percentage of fraud actually caught) and does not add friction for honest customers.

ReturnGuard

Technical Approach

Stack: Python backend, merchant order/return data integration, a risk-scoring model (starts rule-based, can upgrade to ML later), and a simple merchant dashboard. Four key pieces: (1) order/return data ingestion, (2) risk-scoring engine, (3) merchant dashboard showing flagged returns and savings, and (4) auto-approve/flag decision logic wired into the merchant's return flow.

Team & Build Plan

One team member builds the merchant data integration first. Another builds the risk-scoring logic in parallel using a small sample dataset. Once both are working, we connect them into a basic dashboard (built third), then test on one pilot merchant's real historical data before demo.

Impact & Edge

Merchants keep more revenue without punishing honest customers with strict return policies. Unlike enterprise fraud tools priced for large retailers, this is built cheap and specific to Indian COD/return patterns—a gap the project aims to address for small D2C brands.